Virtualisation & Multitenancy

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Outline

Overview

Examples

Virtualisation Approaches

Containers

Multitenancy

Virtualisation

Definition

Enables a **single physical** infrastructure to function as **multiple logical** infrastructure or resource

Many OSs can run and share a single hardware (one-to-many improves utilisation)

E.g., processor, memory, storage, network

Why?

Improve resource utilisation through sharing – lower cost, improve scalability, etc.

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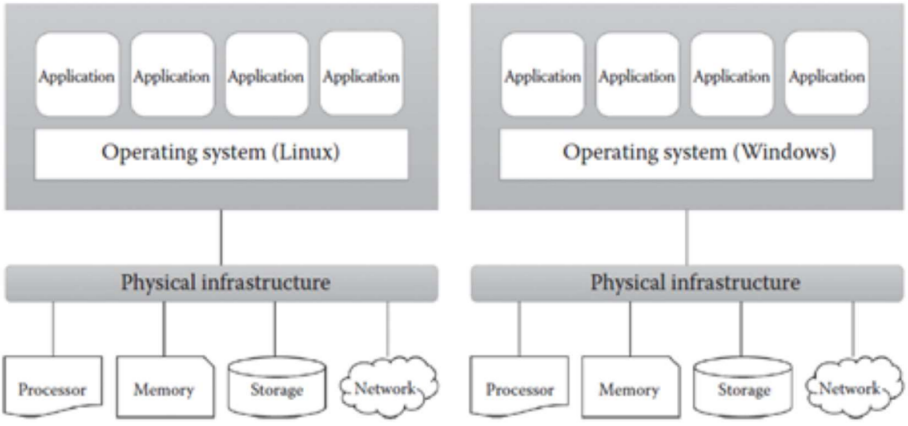
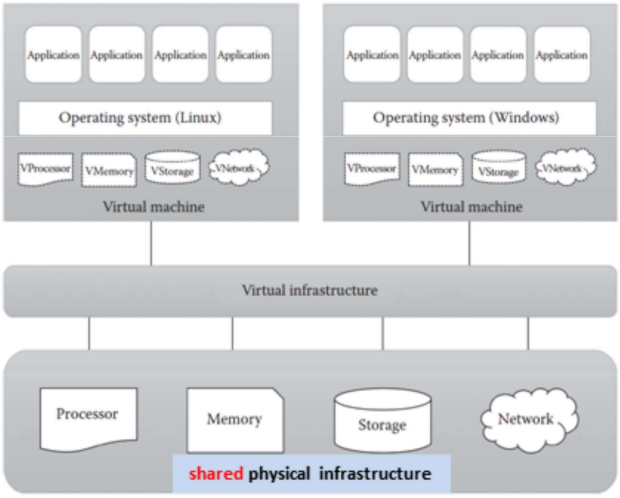
Before Virtualisation



Physical Infrastructure

1:1 relationship with physical infra

After Virtualisation



Virtual Infrastructure

many:1

Virtualisation

Pros

Hardware Independence – software-hardware dependencies in traditional non-virtualised environment

Resource Consolidation – increase hardware utilisation, load balancing, ...

Resource Replication – rapid scaling as resources are virtual (created as virtual disk images)

Cons

Performance overhead

Single point of failure – virtualisation software

Types of Virtualisation

Processor virtualisation

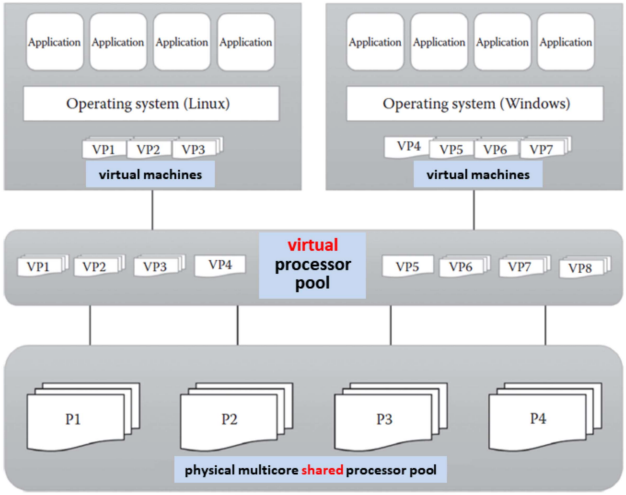
Memory virtualisation

Storage virtualisation

Network virtualisation

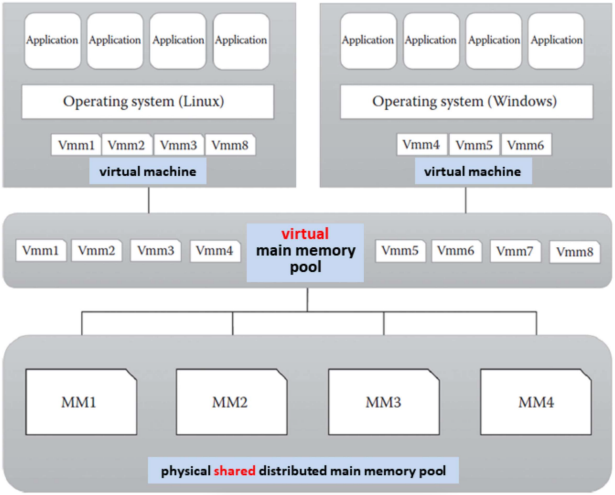
Processor Virtualisation

Abstracting physical processor to pool of virtual processors



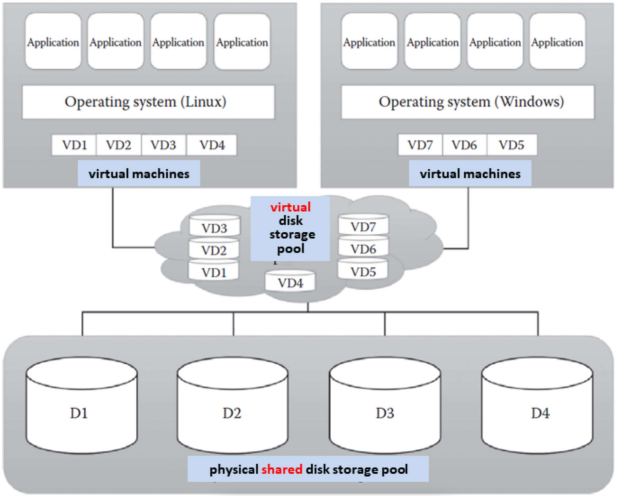
Memory Virtualisation

Providing virtual main memory to the VMs that are abstracted from the physical main memory



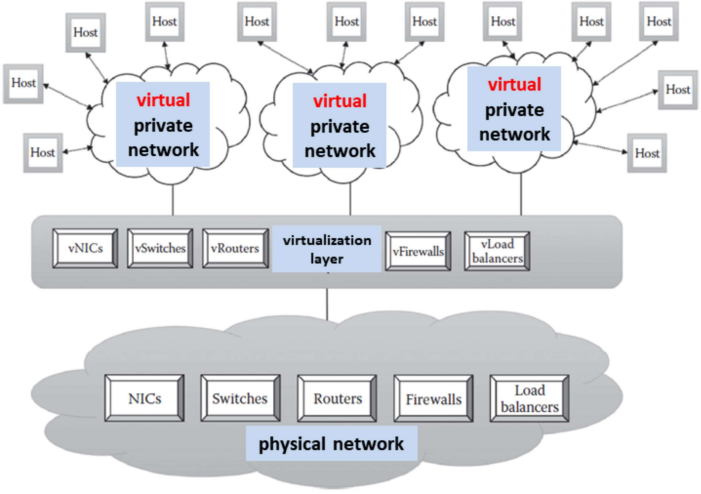
Storage Virtualisation

Providing multiple logical storage that is abstracted from multiple physical storage



Network Virtualisation

Process of abstracting physical network components to form a virtual network



Virtualisation Approaches

Virtualisation Software

Hypervisor or **Virtual Machine Manager (VMM)**

- Software that sits between VMs and physical infrastructure
- Allows IaaS provider to manage VMs and execute consumer's guide OS
- Physical infrastructure resources are shared among the consumer's VMs

Virtualisation Approaches

Full Virtualisation, Para Virtualisation, Hardware-Assisted Virtualisation

Depending on the type of virtualisation approach, the virtualisation software and guest OS will run in different privileged levels

how does the guest talk to the hardware?

only privileged instructions (OS instructions) should reach the hardware
lower the ring number higher the privilege

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Full Virtualisation

VM can run different guest OSs

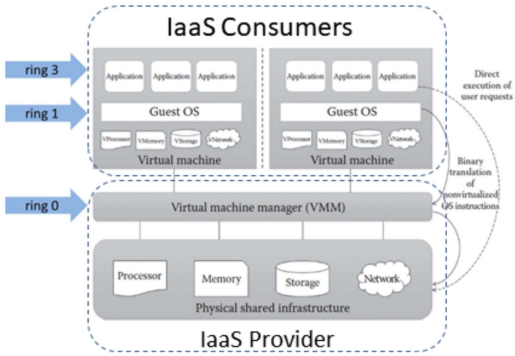
Guest OS is not aware that it is virtualised

VMM at **ring 0** provides all the virtual infrastructures needed for VMs, i.e., guest OS communicates with hardware through VMM

VMM translates guest OS instruction with instructions for the physical hardware

- **Binary translation**

software-assisted



VMM mimics the hardware
Talks to guest OS on hardware's behalf
Then translates the instruction to hardware

Full Virtualisation

Pros

Provides VMs with best isolation and security

Different OSs can run simultaneously

Virtual guest OS can easily be migrated to native hardware

Easy to install and use – no changes to guest OS

Cons

Binary translation overhead – reduced system performance

Need for correct combination of hardware and software – hardware may not have drivers for virtualisation software

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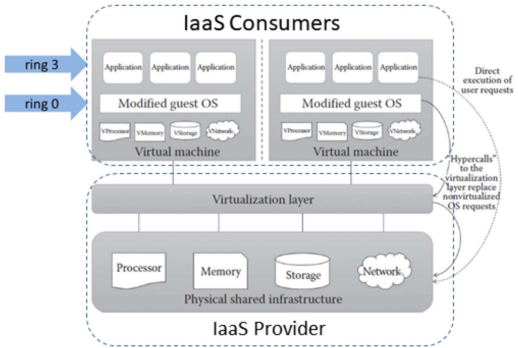
Para Virtualisation

Modified guest OS replaces OS requests with **hypercalls** – also called OS-assisted virtualisation

Hypercalls – system calls with privilege to communicate directly between OS and hypervisor without translation

Guest OS knows it is running in a virtualised environment



OS-assisted

VMM doesnt do translation
It gives guest OS an interface to talk to hardware

Para Virtualisation

Pros

Eliminates overhead of binary translation – improves performance

No need for special hardware

Cons

Overhead of guest OS kernel modifications

Modified guest OS cannot migrate to run on physical hardware

VMs lack backward compatibility, difficult to migrate to other hosts

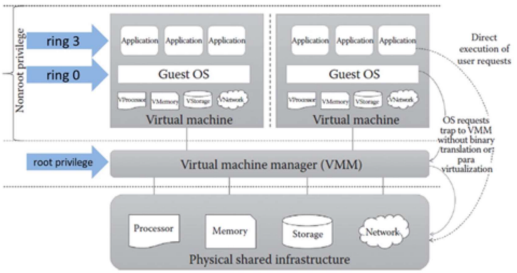
Hardware-Assisted Virtualisation

IaaS Consumer

- Hardware (processor) extension supports virtualisation – guest states are stored in virtual machine control structures
- OS requests directly trap the hypervisor – eliminates binary translation and guest OS modification

VMM has the highest root privilege, while guest OS and user applications have the same non-root privilege

native virtualization



hardware steps in

Virtualisation Approaches

Summary

from perspective of guestOS

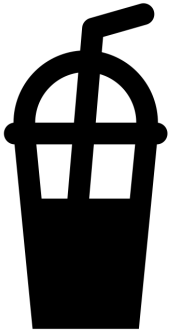
	Full	Para	Hardware-Assisted
Technique	Binary Translation and direct execution	Hypercalls	OS requests trap to VMM without binary translation or para virtualisation
Guest OS modification?	No	Yes	No
Compatibility	Excellent	Poor	Excellent
Is guest OS hypervisor independent?	Yes	No	Yes
Popular vendor(s)	VMware ESX	Xen	Microsoft, Virtual Iron, XenSource

no need guest OS modification but incur translation overhead

avoid overhead but needs to modify guest OS to make hypercalls

best of both worlds

Take a Break



next.. where does virtualization layer sit?

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runs directly on hardware

Type 1

Also called **Bare Metal** (or native) – runs on physical infrastructure without any help from the host OS

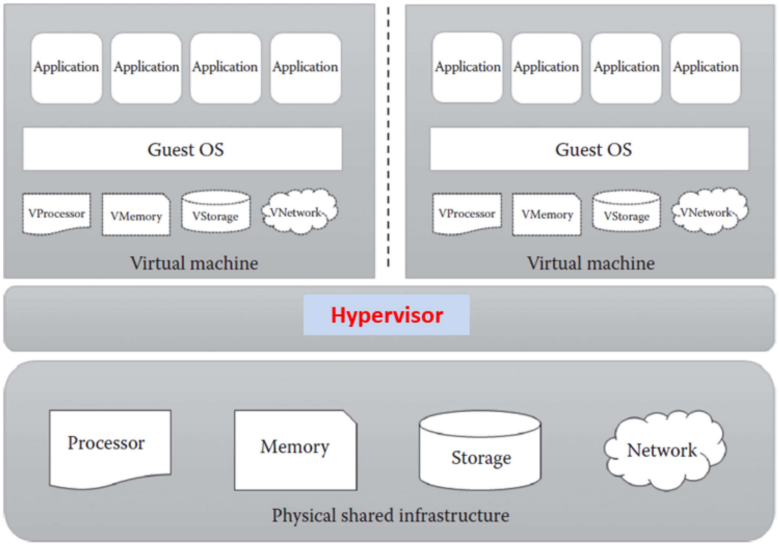
Type 2

Hosted (or embedded) – requires the help of host OS to communicate with underlying infrastructure

e.g. VirtualBox, Parallels…

from perspective of hypervisor

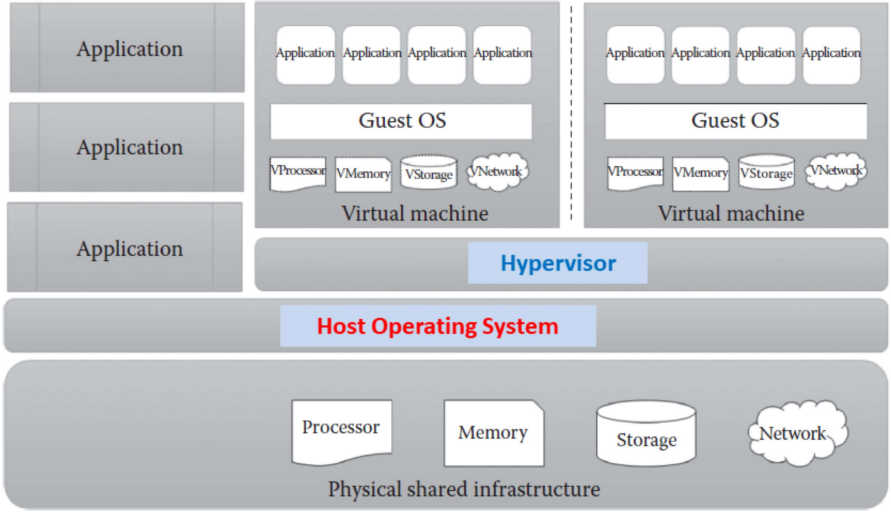
Bare Metal Hypervisor



no host OS

for larger scale applications

Hosted Hypervisor



for smaller applications

Hypervisor	Vendor	Type
Xen	University of Cambridge Computer Laboratory	Type 1
VMWare ESXi	VMware, Inc.	Type 1
Hyper-V	Microsoft	Type 1
KVM	Open virtualization alliance	Type 1
VMWare workstation	VMware, Inc.	Type 2
Oracle Virtualbox	Oracle Corporation	Type 2
Parallels Desktop	Parallels	Type 2

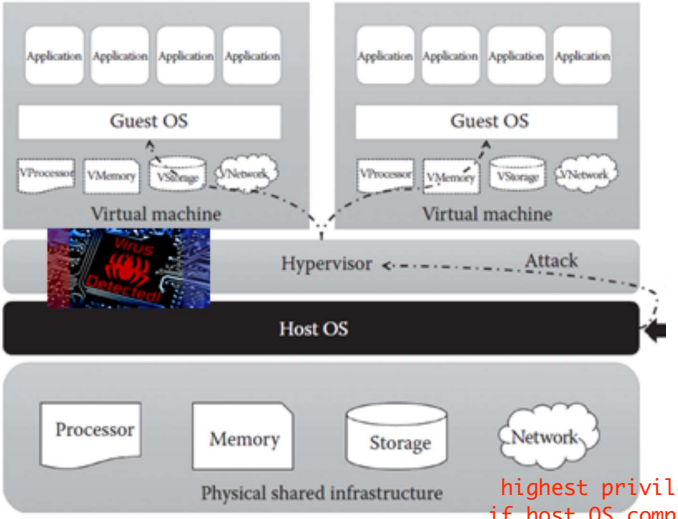
Host OS

Compromise the Host OS

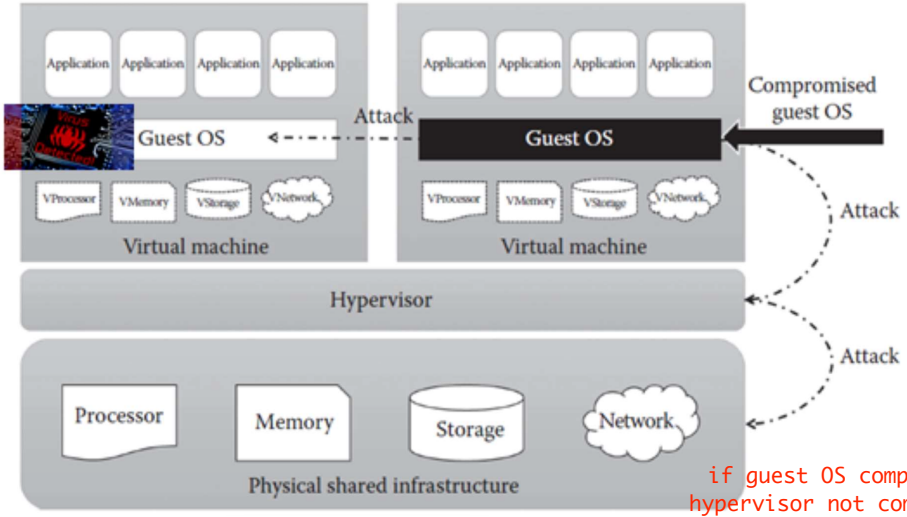
Guest OS

Compromise the Guest OS

`less likely for hypervisor to get compromised`



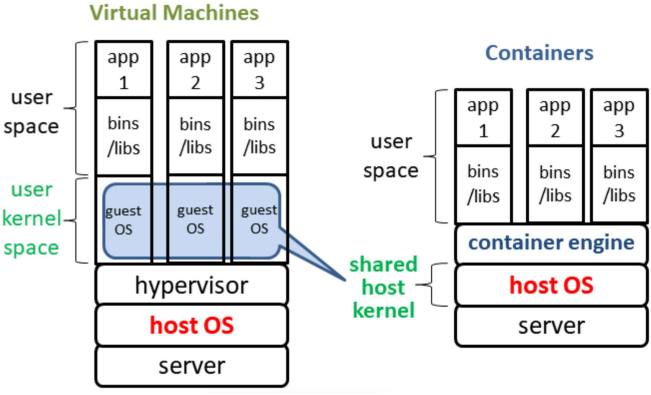
highest privilege -> hypervisor
if host OS compromised hypervisor
can compromise the guest OSes



if guest OS compromised
hypervisor not compromised
virus is isolated

Containers (vs VMs)

do we really need a full fledged VM?



Containers allow existence of multiple isolated user space instances, e.g., Docker

Containers share the host system's kernel with other containers ⇒ lightweight

quickly develop, deploy, test…

What is Docker?

Open Platform for developing, shipping and running applications

Enables you to separate your applications from your infrastructure so that you can deliver software quickly

Ability to package and run an application in a loosely isolated environment called a **container**

Containers are lightweight and contain everything needed to run the application $\Rightarrow$ portability!

Uses the idea of namespaces to provide the isolated workspace

Written in the Go programming language and takes advantage of several features of the Linux kernel

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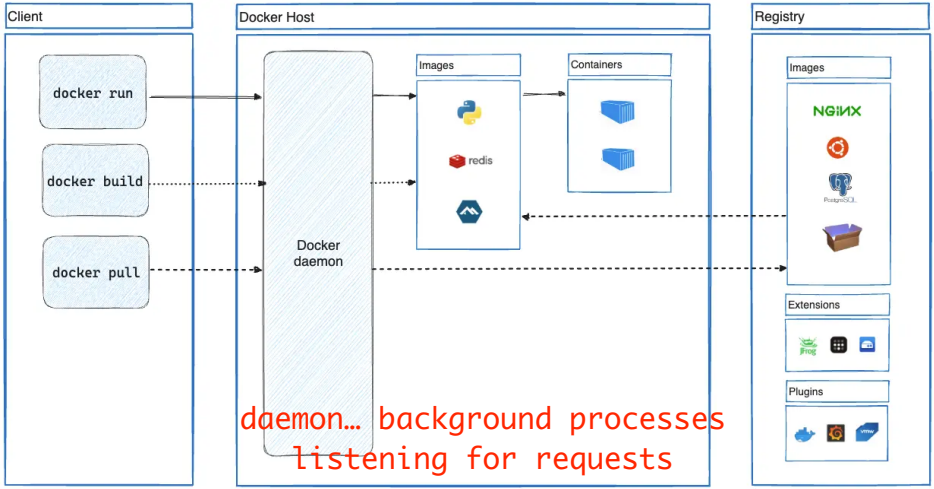
how does Docker run on Mac?… virtualization

Docker Architecture



Uses a client-server architecture

`https://docs.docker.com/get-started/docker-overview/`

Docker Components I

Docker Desktop

Easy to install application for your host OS – enables you to build and share containerised applications and microservices

Includes Docker daemon, Docker client, Docker compose, ...

Docker Components II

Docker Client

Primary way users interact with Docker

Can communicate with more than one daemons

Docker daemon

Building, running and distributing the Docker containers

Listens for Docker API requests

Manages Docker objects such as images, containers, networks, etc.

Can also communicate with other daemons to manage Docker services

Docker client and daemon communicate using a REST API, over UNIX sockets or a network interface

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Docker Components III

Docker registries

Stores images

Docker Hub is a public registry that anyone can use, and Docker looks for images on Docker Hub by default

Images

Read-only template with instructions for creating a Docker container

Often based on another image, but you can also create your own using a Dockerfile

Docker Components IV

Containers

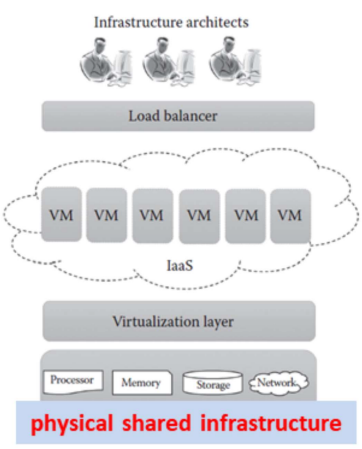
Runnable instance of an image

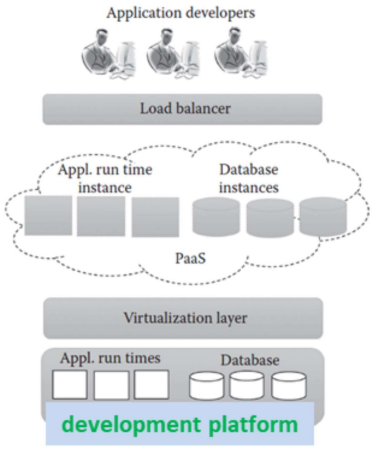
Can create, start, stop, move, or delete a container using the Docker API

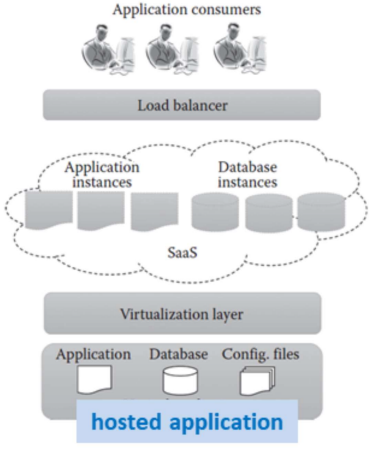
Can control how isolated a container's elements are from other containers or the host machine

`super lightweight`

Virtualisation and Cloud Service Models







IaaS **PaaS** **SaaS**

VMs docker multitenancy

Multitenancy

Cloud Providers

A single instance of *software* runs on a server and serves multiple tenants

Enable multiple users (tenants) to share the same application, e.g., cloud platforms, cloud services, etc.

`e.g. Dropbox, email`

Cloud Consumers

Dedicated instance of software with customised view

Tenants isolation ensures that tenants do not have access to data and configuration information that it does not own

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Multitenancy vs Virtualisation

Multitenancy

Dedicated consumer instance of software

Virtualisation

Abstraction of physical resources

In multitenancy, giving consumers an instance of the software (tenant usage & data isolation) is possible *because* of virtualisation

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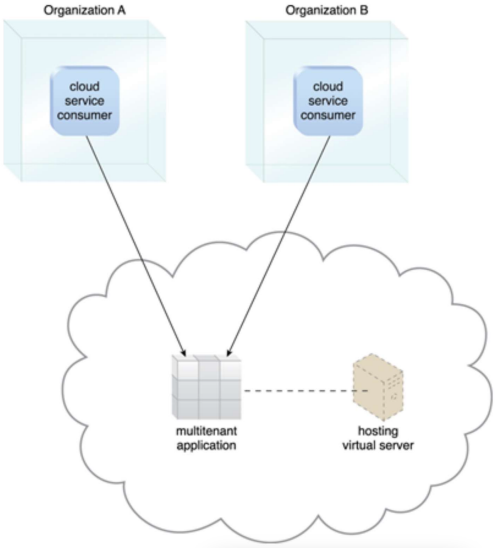
Multitenancy

Example



Multitenancy

Characteristics

Usage/Tenant Isolation – one tenant's behaviour does not affect other tenants

Data Security – separate procedures for each tenant

Recovery – backup, restore

Scalability – usage by tenants and tenant numbers can increase

Metered Usage – each tenant is charged only for features being used

Data Tier Isolation – databases, tables, etc., can be isolated or allowed as shared resources among tenants

different tenants will have different agreements, think Dropbox
e.g. you have 500mb, I have 1gb, someone has 1Tb, etc…

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Next Week

Applications and Paradigms (Part 1)

References

Chapter 5 and Appendix D, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Chapter 7, Essentials of Cloud Computing, CRC Press, 2015

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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